# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME: CSA07 COURSE CODE: Computer Networks**

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**COURSE OUTCOME COVERED**

**CO1:** Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

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Annexure A

**CONCEPT MAPPING**

*Code of Conduct:*

*I* LABHANJA PRAVEEN / 192426007 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:*

| Criteria | Weightage (%) | Q1 10marks | Q2 10marks | Q3 10marks |
|---|---|---|---|---|
| **Concept Identification** | 20% | 2 | 2 | 2 |
| **Concept Mapping Structure** | 25% | 2 | 2 | 2 |
| **Linking & Relationships** | 20% | 1 | 1 | 1 |
| **Application & Analysis** | 20% | 1 | 1 | 1 |
| **Explanation & Clarity** | 15% | 1 | 1 | 1 |

**Total Marks** 21 /30

***Signature of the Faculty***

**CONCEPT MAPPING QUESTIONS**

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

## RUBRICS FOR CALCULATION

| Criteria | Weightage (%) | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| **Concept Identification** | 20% | Accurately identifies all relevant OSI layers, concepts, and relationships | Identifies most concepts with minor omissions | Identifies limited concepts; some inaccuracies | Unable to identify key concepts correctly |
| **Concept Mapping Structure** | 25% | Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections | Concept map is mostly clear with minor structural issues | Concept map lacks clarity, organization, or proper flow | No clear structure or incorrect mapping |
| **Linking & Relationships** | 20% | Clearly establishes correct relationships between layers, functions, and processes | Relationships are mostly correct with minor errors | Limited or partially correct relationships | Incorrect or missing relationships |

| Criteria | Weightage (%) | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| **Application & Analysis** | 20% | Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning | The application is correct but lacks depth in analysis | Basic application with weak explanation | Unable to apply concepts correctly |
| **Explanation & Clarity** | 15% | Provides a clear, concise, and well-structured explanation supporting the concept map | Explanation is understandable with minor gaps | Explanation is unclear or incomplete | No clear or meaningful explanation |

# ERROR CONTROL CONCEPT MAP

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Course: Computer Networks
Code : CSA0710

## Data Link Layer Error Detection

- → Uses CRC (cyclic Redendancy check) or Checksum to Detect errer in frame.
- → Identifies corrupted or lost Data at the link clevel.

## Transport Layer Reliable Deleviery

- → Uses ACK (Acknowledgement) and retransmission
- → Detect Missing or corrpoted Data segments.
- → ensure End-to-End reliable communication.

## Reliable Data Transmission

### Analysis

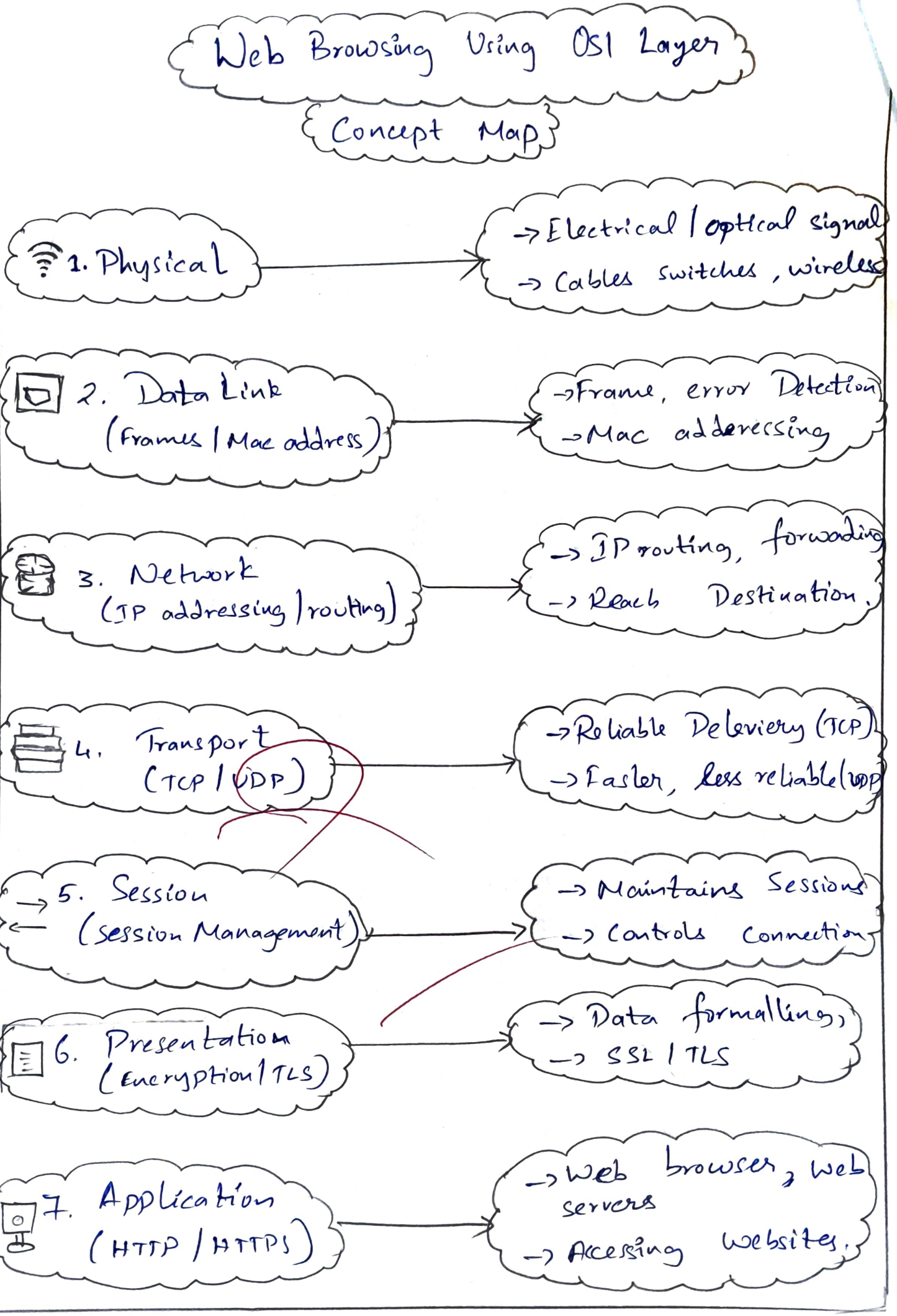
- → The Data Link Layer Detets error in frames using CRC or checksum.
- → The Transport Layer (eg TCP) Detectissing or corrupted Data and can retransmit.
- → Both Layer work togethor to reduce transmission error and ensure reliable Data Deleviely.

### Conclusion

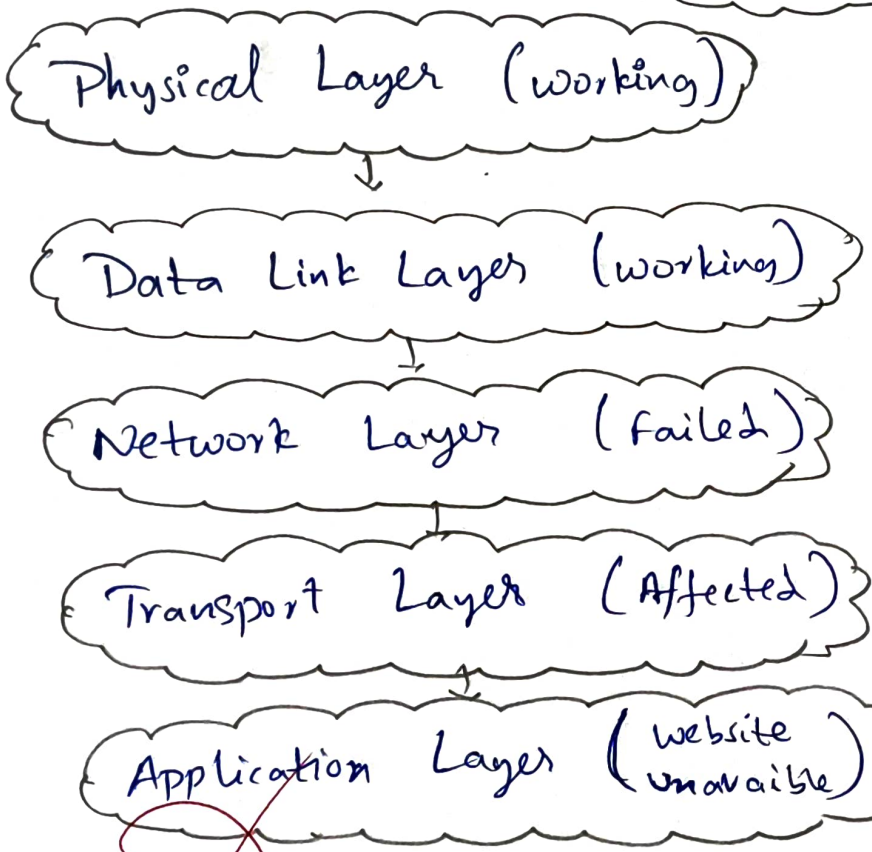
Error Detection identifies problems while transport level recovery helps Delvir the correct Data.

# Web Browsing Using OSI Layer

## Concept Map



# Network Troubleshooting Concept Map




## Explanation

1. Physical & Data Link Layers are working (connection exists and Local communication is fine)
2. Network Layer has failed (IP / route)
3. Transport and Application Layer are affected because they depend on the Network Layer.

## How the concept Map Layer:

- → Identifies the first faulty Layer
- → Avoid check every Layer uncessarily
- → Saves time and take Network repair easier.

## Possible Causes

- → incorrect IP address
- → Wrong gateway
- → Route problem
- → Network configuration issue